George Sideris

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Education

McGill University | Montreal, QC, Canada Aug. 2024 – Present

Master of Science, Mechanical Engineering (Thesis)

- *Graduate Coursework:* Applied Robotics; Fundamentals of Computer Vision; Reinforcement Learning (2nd highest class grade award); Advanced Topics in Vision and Robotics
- *Teaching Assistantships:* Applied Electronics and Instrumentation (F2024); System Dynamics and Control (W2025); Engineering Economics (F2024, W2025, S2025, F2025, W2026)

McGill University | Montreal, QC, Canada Aug. 2019 – May 2024

Bachelor of Engineering, Mechanical Engineering (Internship Program)

- *Graduate Coursework:* Applied Machine Learning; Probability, Statistics and Machine Learning for Mechanical Engineering; Control Systems

Academic Research Experience

Aerospace Mechatronics Lab | McGill University, Canada Aug. 2024 – Present

Graduate Research Assistant (Advisor: Prof. Inna Sharf)

- Developed a high-fidelity Isaac Lab crane-log-pile parallelized simulation environment. Trained a PointNet-based grasp targeting policy via behavior cloning from a privileged-state heuristic and fine-tuned with PPO using an asymmetric actor-critic, achieving 98.2% pile clearing from raw point clouds. Manuscript submitted to IROS 2026. Currently transitioning to real-world deployment on a crane testbed.

NeuroImaging and Surgical Technologies Lab | McGill University, Canada Aug. 2023 – Sep. 2025

Undergraduate Research Assistant (Advisor: Prof. Louis Collins)

- Led a team of four students to create a multi-camera optical tracker for occlusion-robust pose estimation of surgical tool tip with sub-millimeter accuracy. Co-designed and fabricated fixture for attaching cameras to operating room light. Co-implemented multi-camera calibration and object tracking using ArUco fiducials. Designed and implemented an adaptive Kalman filter-based fusion strategy. Achieved 2nd place Engine Capstone Design Prize for Entrepreneurship. Manuscript submitted to IEEE Transactions on Robotics.

Aerospace Mechatronics Lab | McGill University, Canada May 2023 – Aug. 2023

Undergraduate Research Assistant (Advisor: Prof. Inna Sharf)

- Awarded SURE to assist in research towards autonomy of timber-harvesting machines. Created datasets to train a CNN for grapple and log segmentation, grapple opening prediction, and log counting. Integrated draw wire sensor on grapple to experimentally validate CNN. Designed experimental setup emulating log-loader manipulator for anti-sway trajectory planner validation. Manuscript published in ICRA25.

Publications & Manuscripts Under Review

- **G. Sideris**, E. Ayoub, N. Lemieux, H. Fernando, I. Sharf, "Clearing Dense Log Piles with Imitation-Bootstrapped Reinforcement Learning." *Submitted to IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026.*
- **G. Sideris**, J. Cree, A. Stirling, M. Ly, É. Léger, D. L. Collins, "LARK: A Low-Cost, Accurate, Occlusion-Resilient, Kalman Filter-Assisted Tracking System for Image-Guided Surgery." *Submitted to IEEE Transactions on Robotics.* Preprint available at Research Square, doi: 10.21203/rs.3.rs-7622287/v1.
- I. Jebellat, **G. Sideris**, R. Saif and I. Sharf, "Designing Experimental Setup Emulating Log-Loader Manipulator and Implementing Anti-Sway Trajectory Planner," *2025 IEEE International Conference on Robotics and Automation (ICRA)*, Atlanta, GA, USA, 2025, pp. 1488-1494, doi: 10.1109/ICRA55743.2025.11128481.

Industry Experience

OPSIS | Montreal, QC, Canada May 2024 – Present

Co-Founder

- Developing a low-cost, occlusion-resilient surgical tool tracking system for minimally invasive guided surgery. Completed McGill Engine's TechAccel experiential learning program. Awarded Ian McLachlin Prize for Entrepreneurship by McGill Engine. Supervised a business intern during summer 2025, sponsored by McGill Engine.

Lockheed Martin CES | Montreal, QC, Canada Jan. 2022 – Aug. 2022
Jet Engine MRO Engineering Intern (Supervisor: Dr. Fethi Hamdani)

- Created dataset of 100+ labelled images from defective shop floor parts using stereo-microscope. Drafted AI integration sections for $8M CAD NGen proposal on automated visual inspection of HPT parts. Designed tools for jet engine maintenance tasks using AutoCAD; produced drawings and procured materials for 10+ tools. Dispositioned 12 cases of defects on engine parts, saving over $150K CAD.

Bombardier Aerospace | Montreal, QC, Canada May 2021 – Dec. 2021
Project Engineering Intern (Supervisor: Ryan Jayatunge)

- Tracked stress and materials & processes items from inception to disposition using Power BI. Developed a data tracking tool for managing aircraft design modifications using VBA, reducing weekly workload by 4 person-hours.

Community Involvement & Leadership

McGill Vertical Flight Society Projects Team | McGill University, Canada Aug. 2020 – June 2023
Co-Captain / Captain / Structures Lead

- *Team Co-Captain (Aug. 2022 – June 2023):* Co-led a team of twenty students to design and build a 20 lb., 10 ft dia. blended-wing tail sitter that can perform hover to cruise transition. Gave presentations on UAV design and manufacturing and held workshops for technical skills (digital electronics, Arduino programming, composites manufacturing). Achieved 3rd place at the 2023 DBVF competition and received Perseverance award.
- *Team Captain (Aug. 2021 – July 2022):* Led a team of eight to design and build a 15 lb., 6.5 ft dia. quadrotor with airfoils capable of hover-to-cruise transition. Designed custom frame parts, performed FEA using Abaqus, and developed wing-tilt mechanism with Arduino-based control. Received Best Drawing Package award.
- *Structures Lead (Aug. 2020 – July 2021):* Collaborated with eight teammates to design and manufacture a 1 m dia. hex-copter. Led a sub-team to create the team's first custom frame. Designed frame CAD models and drawings in SolidWorks.

McGill Institute for Aerospace Engineering | McGill University, Canada June 2021 – June 2022
VP External

- Organized inaugural AeroFair, a job fair with recruiters from 6 aerospace companies, attended by 100+ students. Organized two Python workshops: Intro to Git/APIs and CNNs for Fault Classification.

Canadian Robotics Competition (CRC) Teams | Montreal/Laval, QC, Canada Sep. 2016 – May 2019
Captain / Team Member

- Captain, Marianopolis Robotics Team (2018–2019); Member, Marianopolis Robotics Team (2017–2018); Member, Laval Senior Academy Robotics Team (2016–2017).

Presentations (Institutional)

- *Capstone Poster Presentation, 2024/04/03, McGill University.* Poster Title: Multi-Camera Surgical Tool Localization Setup for Image-Guided Neurosurgery
- *Undergraduate Research Poster Presentation, 2023/08/17, McGill University.* Poster Title: Dataset Creation and Experimental Validation of a Multi-Stage CNN for Log Loader Grapple Characterization
- *McGill Institute for Aerospace Engineering Internship Panel, 2022/10/25, McGill University.* Presentation Title: My Internship Experiences in Montreal's Aerospace Sector

Honours & Awards

- *FRQNT Master's Training Scholarship (2025/06 – 2026/12)*: $26 667 provincial graduate research scholarship.
- *Mitacs Accelerate Research Award (2025/11 – 2026/06)*: $30,000 graduate research internship in collaboration with FPInnovations, focusing on autonomous log-pile pick-and-place and industrial robotics solutions.
- *Ian McLachlin Prize for Entrepreneurship in Engineering (2024/11)*: Awarded to high-achieving engineering students who have started or advanced an entrepreneurial project with business or social impact.
- *Engine Capstone Design Prize (2024/04)*: Recognition of teams with innovative final-year projects with startup potential.
- *SURE Award (2023/05 – 2023/08)*: $9,000 undergraduate research traineeship.
- *VFS DBVF Competition 3rd Place Team Award (2023/06)* (Role: Co-Captain).
- *VFS DBVF Competition Best Drawing Package Distinction (2021/06)* (Role: Structures Lead).
- *McGill Science Prize (2019/08)*: Recognition of student achievement and promise in science and mathematics.

- *Silver Key Scholarship (2018/10)*: $1,500 scholarship for academic achievement and community involvement.

Academic Service

- **Peer Reviewer**: IEEE International Conference on Robotics and Automation (ICRA 2026); IEEE Robotics and Automation Letters (RA-L 2026).

Technical Skills

- **Programming**: Python, C++, ROS, Linux (Unix Shell), Isaac Sim, Isaac Lab, Matlab, Simulink, VBA, Excel, LaTeX
- **Design**: SolidWorks, Fusion 360, AutoCAD, Abaqus, Mastercam, Cura